

Module VI
Single Phase Induction Motors

- Single-phase induction motors
- double revolving field theory
- torque slip characteristics
- types of Single-phase induction motors
- Split phase, capacitor-start induction-run,Permanent capacitor types
- applications

3. Single-phase Induction Motor

- Single phase induction motors are generally used for applications which require capacity less than 0.5kW
- Merit: Simple construction, reliable, easy to repair, cheaper.
- Demerit: Low overload capacity, low efficiency, low power factor, low output as compared to 3-phase induction motor.

Construction

- Rotor is same as that of 3-phase squirrel cage induction motor.
- Single phase winding in the stator.
- Winding is usually concentric coils
- **Main winding or Running winding** and **Auxiliary winding or Starting winding**
- Centrifugal switch cut-out auxiliary(starting) winding after starting.

Construction

- **Stator**
- The stator is made up of silicon stamping

(silicon has a property of reducing the hysteresis losses of the motor stampings are in laminated format for reducing the eddy current loss)

- The stampings are slotted in the motor to carry the winding of the motor
- Copper coils (insulated) are used
- The coils are wound in the motor to form a certain number of poles.
- **Rotor:**
- It is a robust construction.
- The rotors are made up of aluminum or copper bars.
- Two end side of these bars are permanently short-circuited with the help of end rings.
- Such construction is called a squirrel cage rotor.

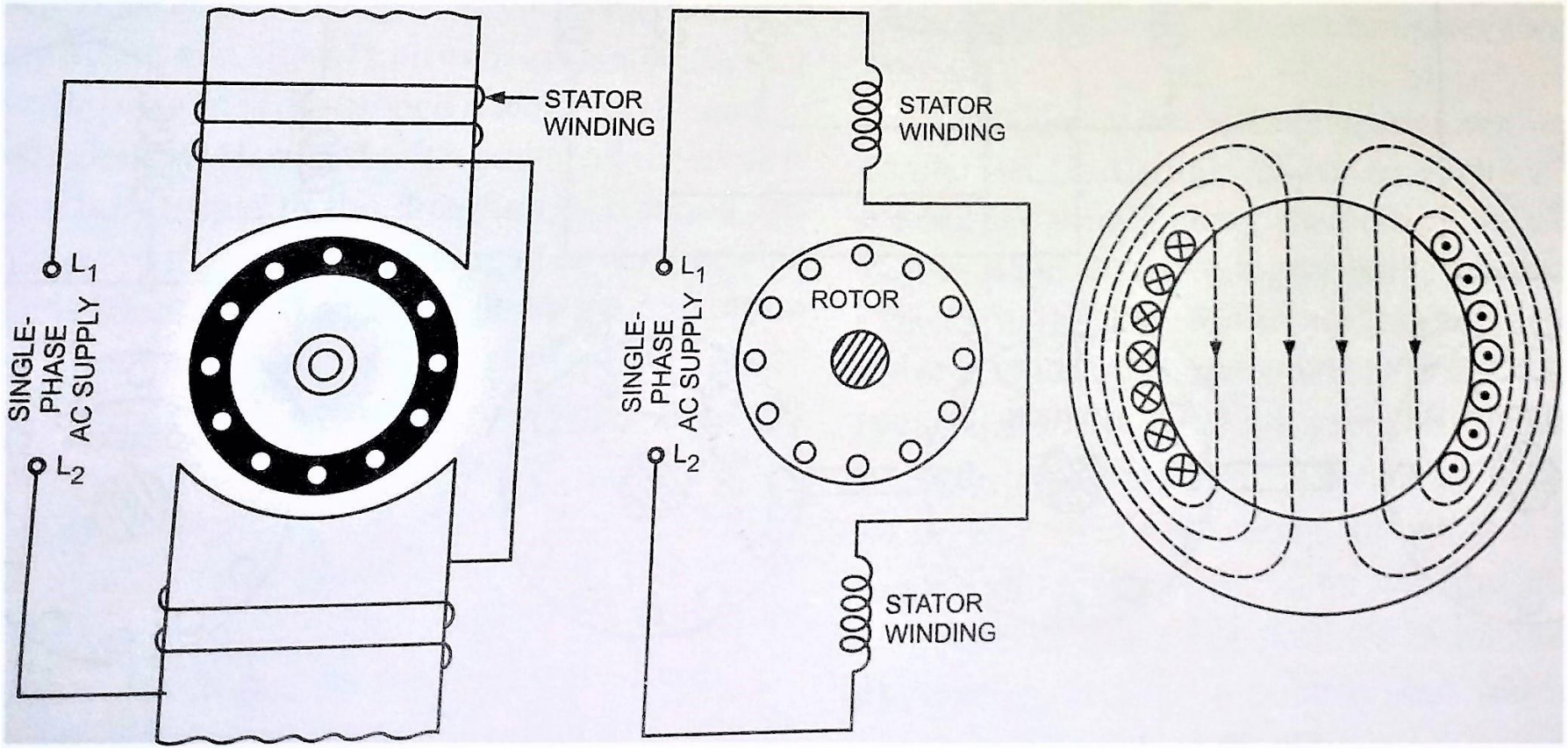


Figure 20 : Single phase induction motor

Principle of Single-phase Induction Motor

- Stator is connected to 1-phase AC supply
- Magnetic field is developed.
- Axis of magnetic field will be always along the axis of stator coils.
- Field is stationary(not rotating) but pulsates in magnitude and varies sinusoidally with time.
- Current is induced in the rotor → axis of rotor mmf coincides with that of stator field → torque angle is zero → no starting torque is developed.
- Single phase induction motor is **not self-starting**
- If the rotor is given a push, it will pick up speed and continue to rotate developing the operating torque.

3.1 Double Revolving Field Theory

Double field revolving and cross-field revolving, two basic theories use to explain why single-phase induction motors are not self-starting,

Pulsating field produced in a single phase motor can be resolved into two components of half it's amplitude & rotating in opposite directions with synchronous speed

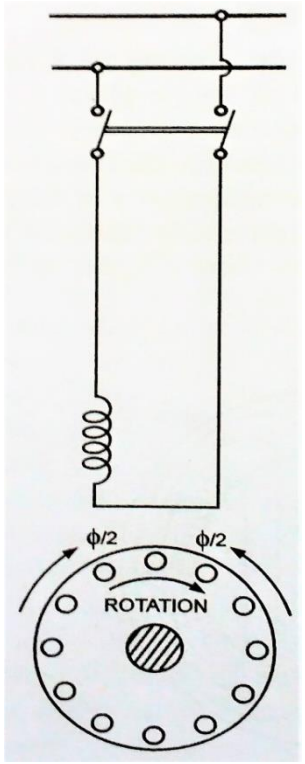


Figure 21 : Circuit(a)

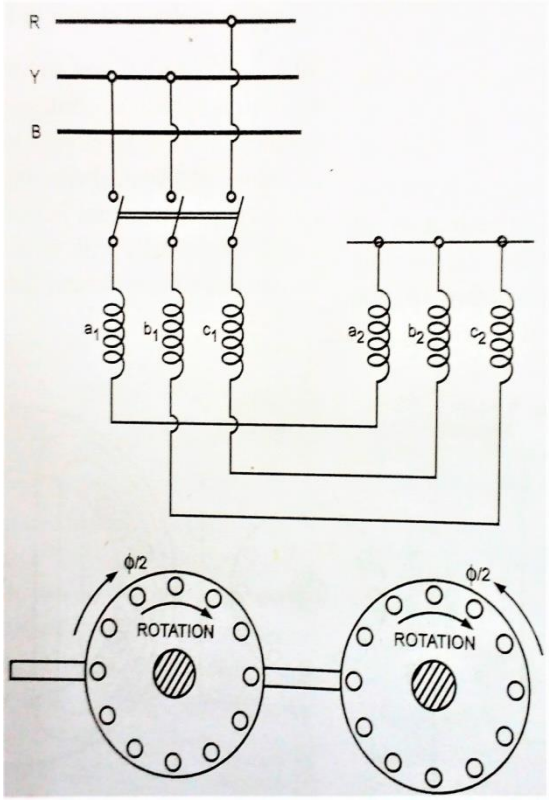


Figure 22 : Circuit(b)

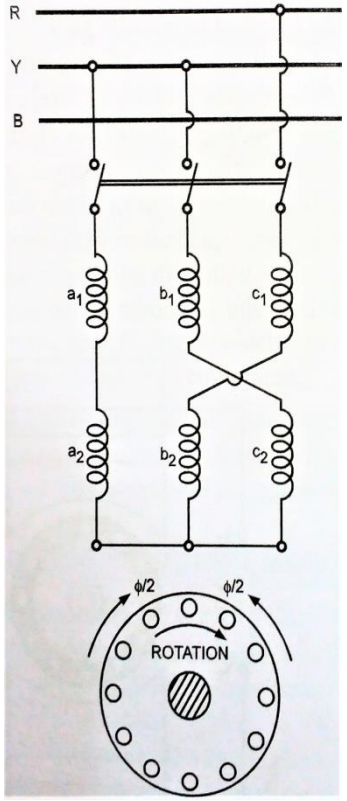


Figure 23 : Circuit(c)

3.1 Double Revolving Field Theory

- At standstill → N = 0 → Slip, s = 1 → Two rotating magnetic field pass the rotor at same slip inducing equal currents in the squirrel cage rotor.
- Two rotating magnetic fields are of same strength → develop equal & opposite electromagnetic torques → Net torque is zero =⇒ Starting torque is zero → Hence **single phase induction motor is not self-starting**.
- If the rotor is made to run at speed 'N' by external means in any direction(eg: in the direction of forward field) → now the slips are 's' & (2-s).
- Slip of rotor with respect to forward rotating field(F_f),

$$s_f = \frac{N_s - N}{N_s} = s$$

- Slip of rotor with respect to the backward rotating field(F_b),

$$s_b = \frac{N_s - (-N)}{N_s} = \frac{2N_s - (N_s - N)}{N_s} = (2 - s)$$

3.1 Double Revolving Field Theory

- During normal operation, (2-s) >> s→ backward field rotor currents are much larger than at standstill & have low power factor → corresponding opposing rotor mmf causes the backward field to be greatly reduced in strength.
- Low speed forward rotating field induces smaller currents of higher power factor in rotor than at standstill → strengthens forward field.
- Weakening of backward field & strengthening of forward field depends upon the slip(or speed of rotor). It increases with decrease in slip w.r.t forward field or increase in rotor speed in forward direction.
- **At near synchronous speed, forward field >> backward field. Hence there is a net running torque.**

3.2 Torque-Slip Curve

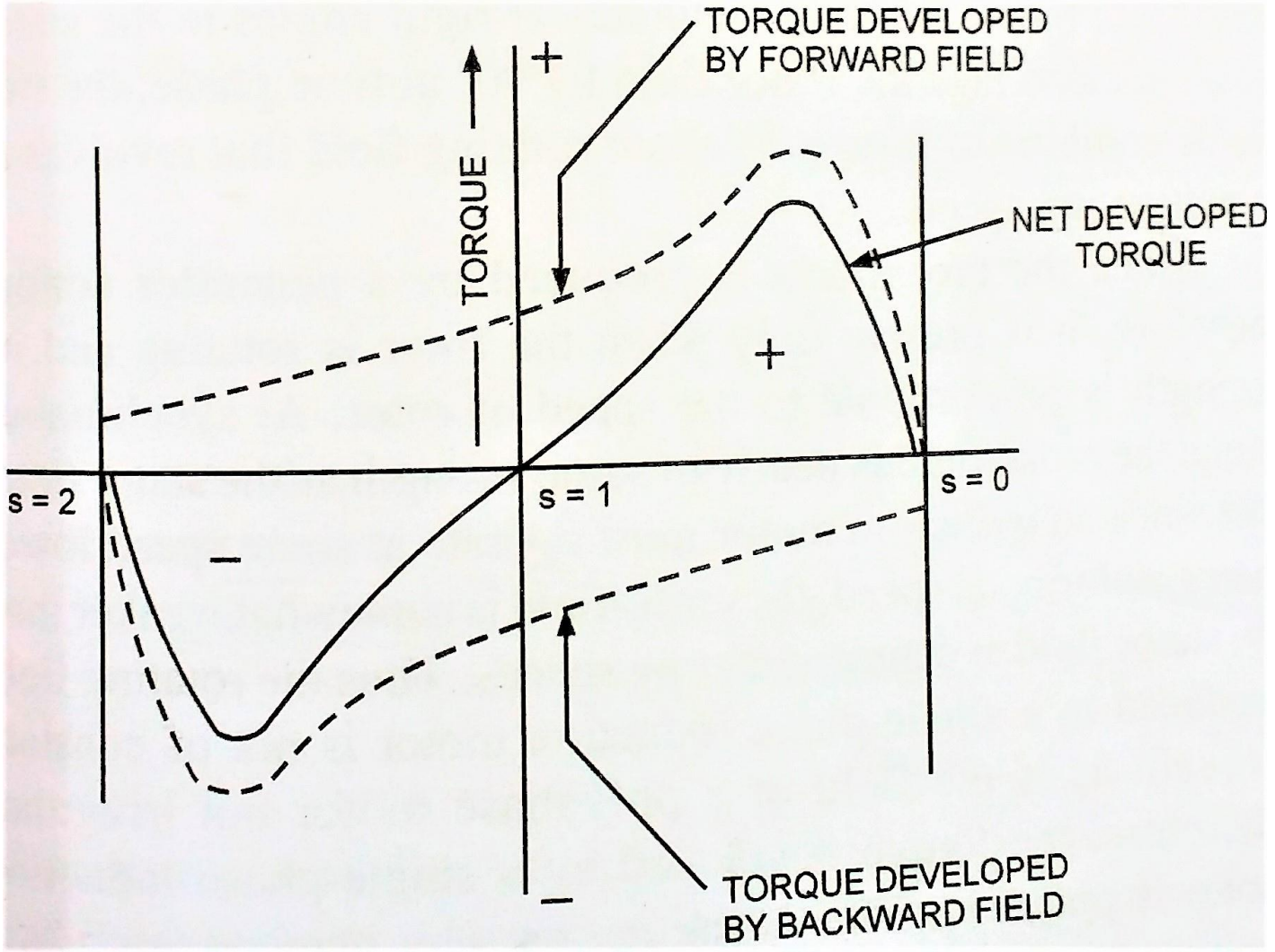


Figure 24 : Torque-Slip curve of Single Phase Induction Motor

3.2 Torque-Slip Curve

- Torque produced by forward field in the direction of rotation → +ve
  Torque produced by backward field → -ve
- Net torque = Algebraic sum of individual torques produced by forward & backward fields.
- **Net torque passes through zero at slip = 1 (at standstill) =⇒ Starting torque = 0 → Motor can't start rotating independently.** For other values of slip, motor develops a net torque in the direction of rotation.
- Maximum torque is also less.
- **Resultant torque drops to zero at a speed slightly below synchronous speed** (whereas in 3-phase induction motor, torque doesn't drop to zero until synchronous speed is attained).
- Net torque at synchronous speed is negative (due to backward field).
- Single phase induction motors are noisier than 3-phase induction motor due to the presence of second harmonic pulsating torques

3.4 Types

- Single phase induction motors are not self-starting. Hence it is necessary to employ some external means for making it self-starting.

Types

1. **Split-Phase Induction Motor** or **Resistance-Start Single-phase Induction Motor**
2. **Capacitor-Start Induction Motor**
3. **Capacitor-Start and Capacitor-Run Induction Motor**
4. Permanent Capacitor Single-phase Induction Motor
5. **Shaded Pole Induction Motor**

Split-Phase Induction Motor

Principle of Phase Split

- If two windings, spaced 90° electrical degrees apart & connected in parallel on the stator of a motor is excited by a single phase source, an alternating field will be produced but it will not rotate.
- If an impedance(resistance, inductance or capacitance) is connected in series with one of the windings, the currents may differ in time phase. By proper selection of impedance, the currents may be made differ by as much as $90^o \rightarrow$ rotating field is produced(as in case of two motor)

$\rightarrow$ starting torque $\rightarrow$ motor will become self-starting.

- Main winding or Running Winding
- Auxiliary winding or Starting winding

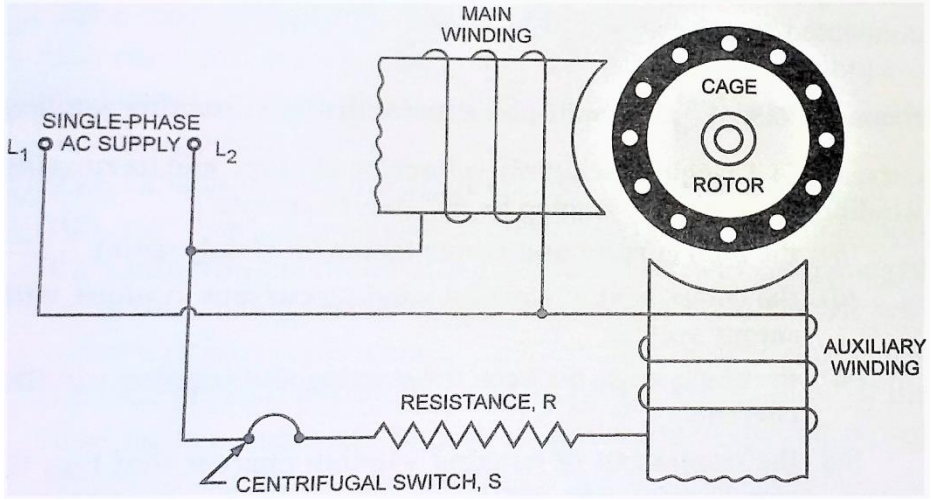


Figure 29 : Split-phase Induction Motor

Split-Phase or Resistance-Start Induction Motor

- High resistance is connected in series with auxiliary winding.
- Main winding → High inductive reactance & Low resistance.
- Auxiliary winding → Low inductive reactance & High resistance.
- Auxiliary or starting winding current(I_s) doesn't lag much supply voltage(V).
- Main winding current(I_m) lags supply voltage(V) more than that of I_s.
- The phase shifted currents in main winding & auxiliary winding produce rotating magnetic field → motor starts.
- Auxiliary winding is removed during running condition by the centrifugal switch.
- Starting torque(T_{st}) = (1.5 to 2) times full-load torque(T_{FL})

  Starting current(I_{st}) = (6 to 8) times full-load current(I_{FL})

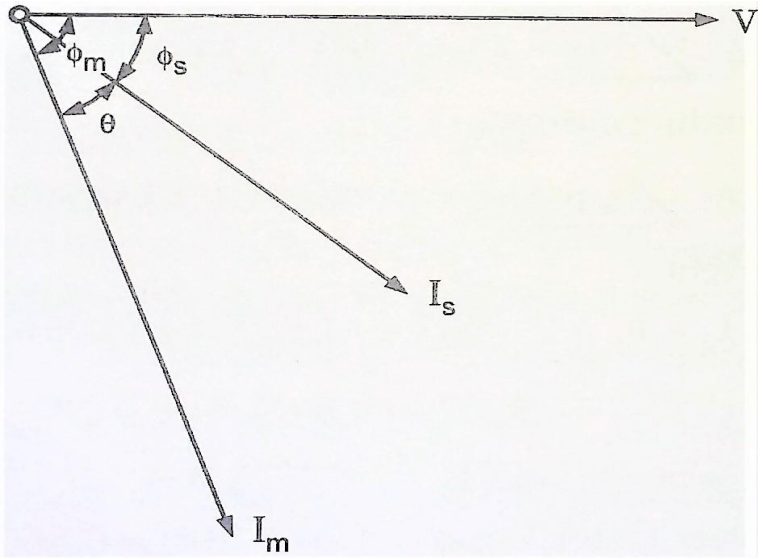
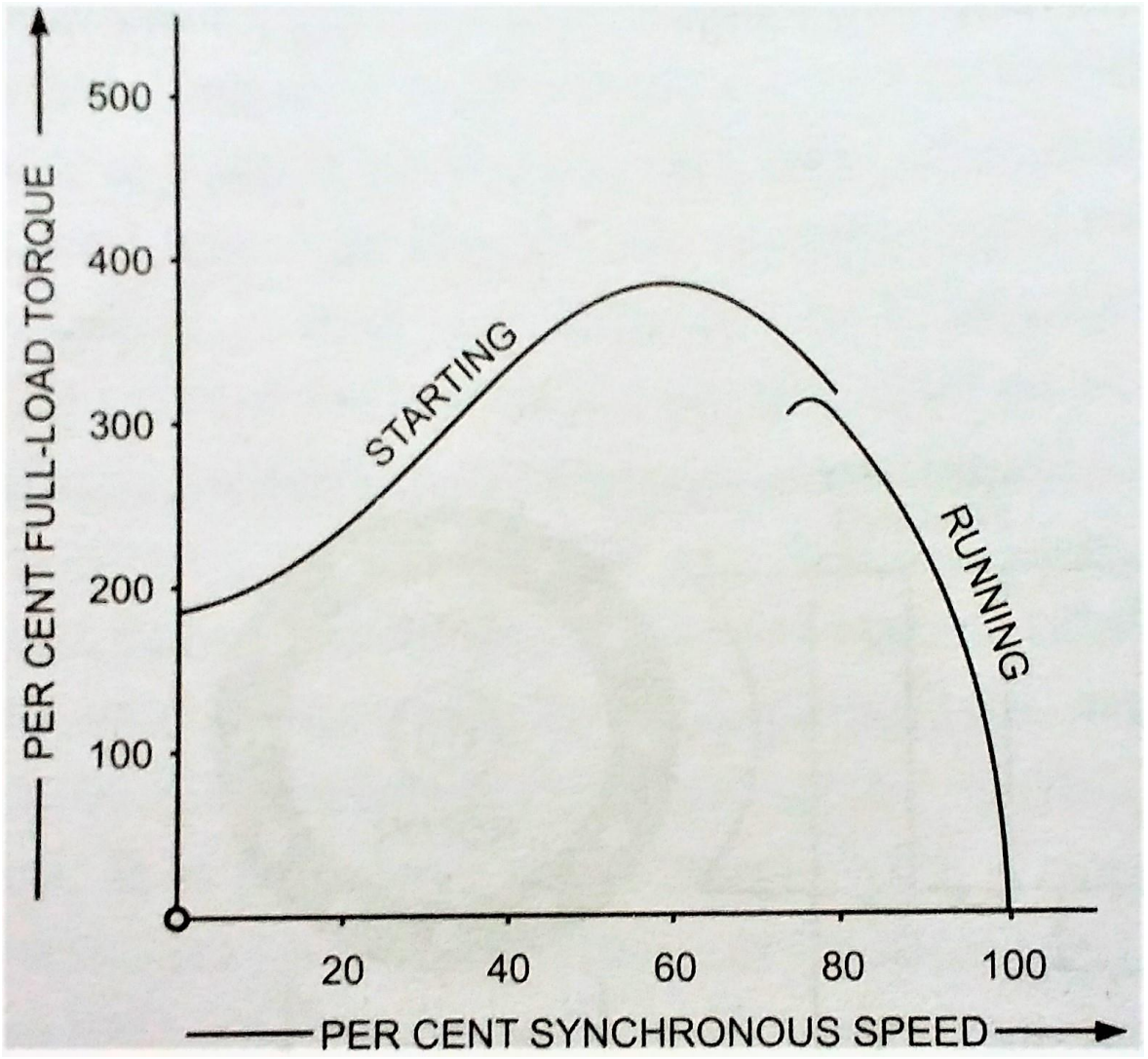


Figure 30 : Phasor Diagram

Split-Phase or Resistance-Start Induction Motor



Capacitor-Start Induction Motor

- Also known as **Capacitor-Start Induction-Run Motor**
- An electrolytic capacitor is connected in series with the auxiliary winding.
- By using a capacitor of proper value, starting winding current(I_s) can be made to lead the main winding current(I_m) by $90^o \Rightarrow \theta = 90^o$.
- Auxiliary winding is removed during running condition by the centrifugal switch.
- Starting torque(T_{st}) = (3.5 to 4.5) times full-load torque(T_{FL})

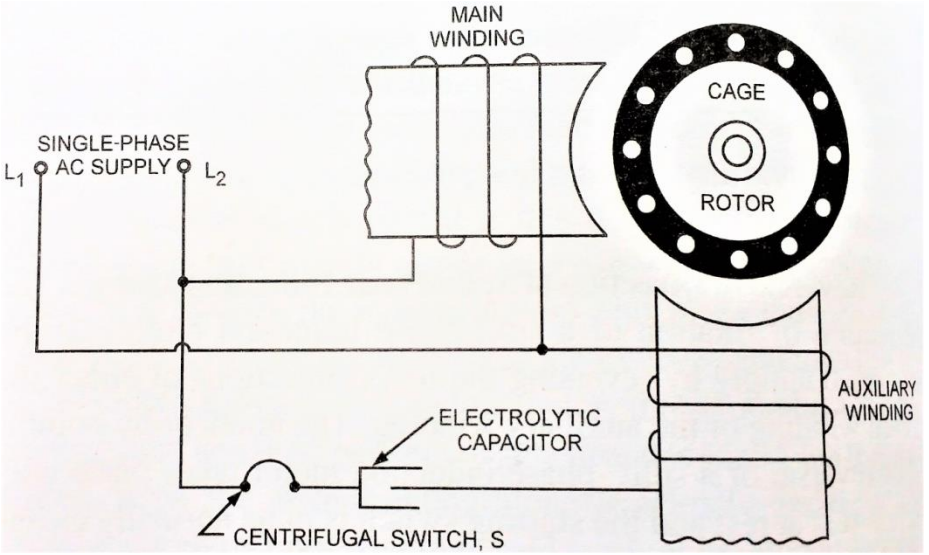


Figure 32 : Capacitor start induction motor

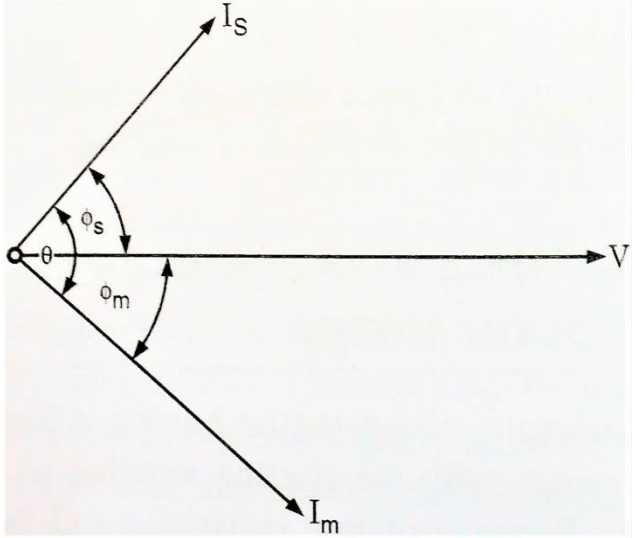


Figure 33 : Phasor diagram

Capacitor-Start and Capacitor-Run Induction Motor

- Auxiliary winding & a capacitor remain connected in the circuit at all times.
- Capacitor employed for normal running(C_R) should be of continuous- duty rating → Small value oil impregnated paper capacitor.
- Capacitor employed for starting(C_S) should be of short-duty rating → Large value electrolytic capacitor.
- When motor attains 75% of synchronous speed, starting capacitor is cut-out from the circuit by the operation of centrifugal switch.
- C_S = (10 -15)times C_R
- Better powerfactor (0.8 -0.95)
- High cost

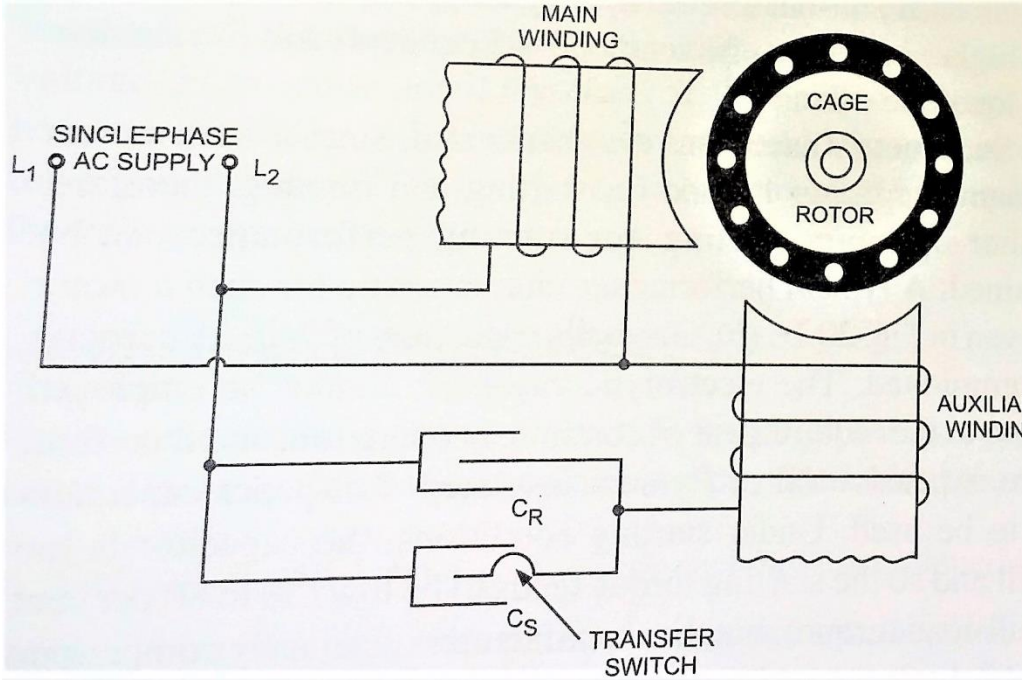


Figure 34 : Capacitor-Start and Capacitor-Run Induction Moto

Shaded Pole Induction Motor

- Stator has salient poles
- A part(about 1/4 to 1/3) of each pole is wrapped by a copper strap forming a closed loop known as **shading coi**
- Stator winding is connected across the ac supply → alternating current flows through the winding
- Change of flux induces short-circuit currents in the shading coil in such a direction as to oppose the core flux
- Flux in the shaded part of the pole lags behind the flux ir the main part of the pole.
- Main flux & shaded pole flux are displaced by approximately 90°

→ rotating magnetic field is set-up → torque is developed.

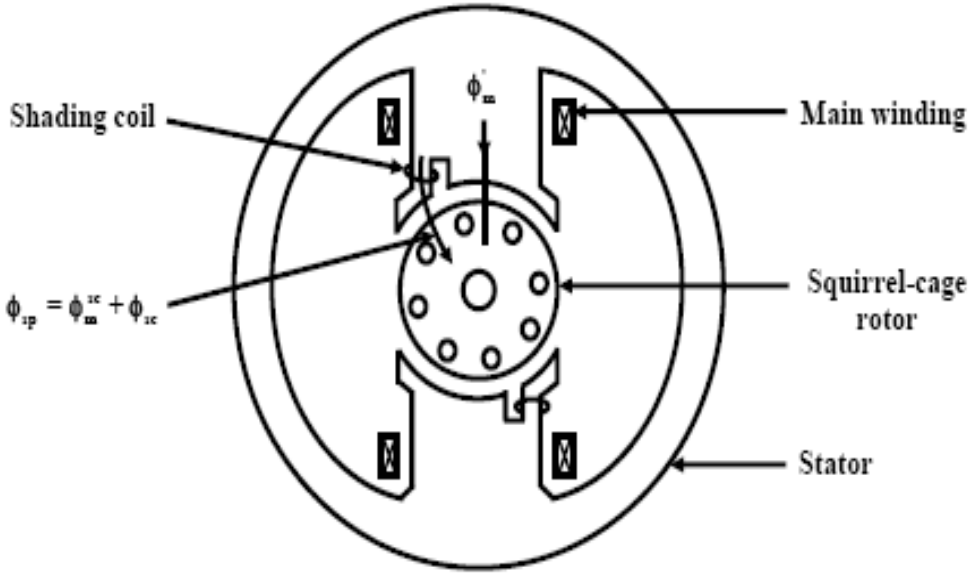


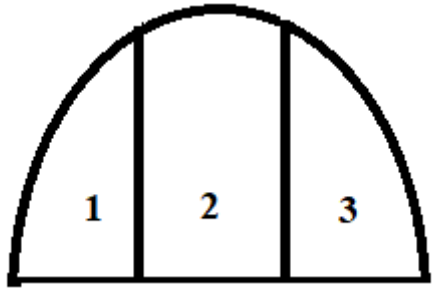
Figure 35 : Shaded Pole Induction Motor

Shaded Pole Induction Motor

Working:

(i) From zero to near positive maximum current

- Current in the winding increases → emf & current are induced in shading coil → shading flux opposes main flux → greater portion of main flux passes through unshaded part of pole.
- Resultant flux lies near the centre of the unshaded part of the pole.



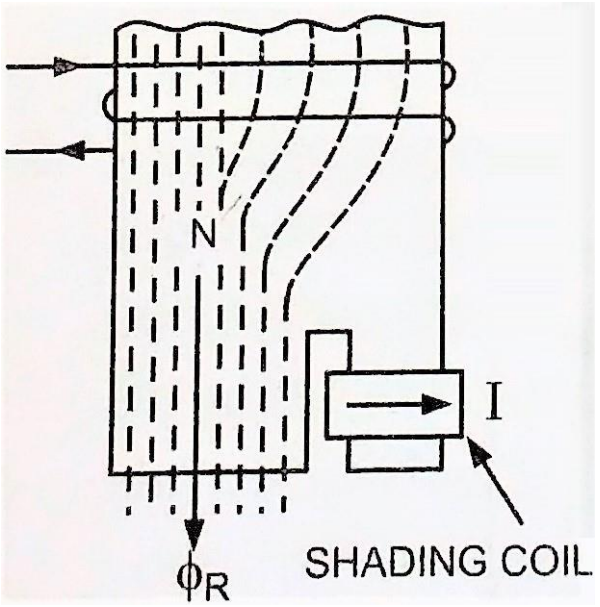


Figure 36 : Shifting of air-gap flux

Shaded Pole Induction Motor

(ii) In the region of maximum current

- Rate of change in current & flux are zero → induced emf & hence current in shading coil becomes zero → flux distribution is uniform over the entire pole face.

- Resultant flux lies at the centre of the pole.

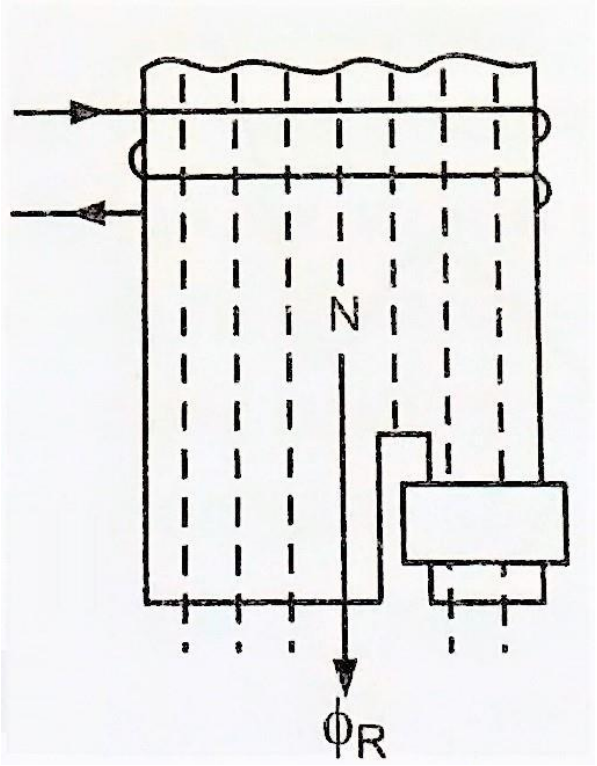


Figure 37 : Shifting of air-gap flux

Shaded Pole Induction Motor

(iii) From nearly positive maximum to zero

- Current in the winding decreases → emf & current are induced in shading coil → shading flux aids main flux → greater portion of main flux passes through shaded part of pole.
- Resultant flux lies at the centre of the shaded part of the pole.

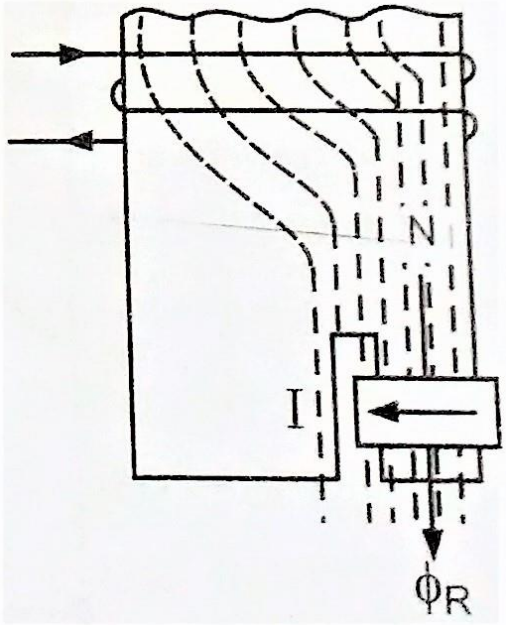


Figure 38 : Shifting of air-gap flux

→ Effect of shading coil is to cause the field flux to shift across the pole face from unshaded portion to shaded portion → Shifting flux is like a rotating weak field moving in the direction from unshaded portion to shade portion of the pole → starting torque is produced → motor starts.

3.5 Applications

Single phase induction motors are widely used in domestic as well as commercial fields.

- Washing machine
- Fans
- Blowers
- Refrigerators
- Grinders
- Centrifugal pumps
- Hair dryers

THANK YOU